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Source: *Ambio*, Vol. 25, No. 5 (Aug., 1996), pp. 314-317

Published by: [Springer](#) on behalf of [Royal Swedish Academy of Sciences](#)

Stable URL: <http://www.jstor.org/stable/4314484>

Accessed: 04/07/2012 06:32

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Destruction and Management of Mount Kenya's Forests

This article presents data on the destruction of the montane forests on Mount Kenya. The material was obtained during fieldwork for a phytosociological study in 1992–1994. Special emphasis was given to the observation of regeneration patterns and succession cycles within the different forest communities, with regard to the impact of humans and big game. Although private tree planting is reducing the fuelwood deficit in Kenya, large parts of the 200 000 ha of Mount Kenya's forests—the largest natural-forest area in the country—are heavily impacted by among other things illegal activities. The wet camphor forests of the south and southeast mountain slopes are being destroyed at an alarming speed, by large-scale selective logging of *Ocotea usambarensis* and marihuana cultivation. The drier *Juniperus procera* forests are also logged, but are even more endangered by the new settlement schemes. The large elephant population does not affect forest regeneration; whereas browsing and chafing by buffaloes inhibits regeneration of the dry forests, and damages many trees. Suggestions are presented for better management of the forest resources.

INTRODUCTION

The forests of Mount Kenya have been subjected to heavy logging for decades. Moreover, due to selective logging, they have also undergone significant qualitative changes. For Kenya, the former natural forest cover is estimated to have been about 80 000 km² or 15 % of the total area of the country (1). Today, even including plantations of exotic species, only a quarter of this area merits the term forest. Two hundred years ago Kenya had 40 000 km² of forests. The trend for deforestation is given in Barnes (2) who gives a figure of 6900 km² for indigenous Kenyan forests in 1980.

Although private tree planting has considerably reduced, e.g. the fuelwood deficit (3), destruction of natural forests has still increased drastically. With respect to a present growth rate of 3.8% of the population and an annual deforestation rate of 1.5%, natural forests will have been extinguished by the year 2040 (2).

In the 1970s, forests were mainly destroyed in order to produce charcoal, of which Kenya used to export about 58 000 tons to the Middle East per year; i.e. 96% of the total charcoal exports of Africa (2). Softwood plantations, especially *Pinus radiata* and *Pinus patula* (4), were then established in a reforestation program. Today, most forest products—mainly timber and firewood—are used within the country.

Vast areas of forest are still being exploited and destroyed by frequent illegal cutting, or are being cleared for new settlement schemes. In the late 1980, 12 000 ha were cleared in the southwestern Mau forests and 15 000 ha in the South Nandi area (5). In late 1993 and early 1994, forest and plantation areas were extensively burned, following a series of presidential decrees, which permitted settlers who were evicted after the abolition of the shamba system in 1987, and new squatters to move into the forests and to slash and burn and finally to cultivate the cleared areas.

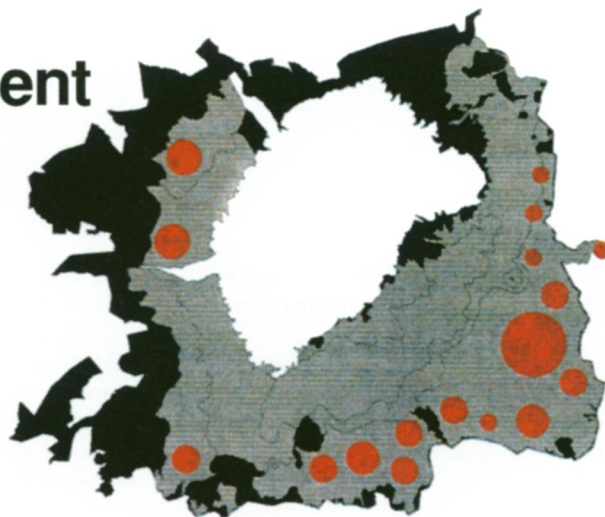


Figure 1. Vegetation changes in the Mount Kenya Forest Reserve after 1963. Destroyed or extremely disturbed areas are shown in black, more or less undisturbed indigenous forests are shown grey, red circles indicate areas of heavy logging. Source: Kenya Indigenous Forest Conservation Project.

FOREST DESTRUCTION AND MANAGEMENT

A total of 40 sawmills were operating in the Mount Kenya area in 1991 (6), all of them well equipped with high-tech logging and sawing equipment. In contrast, none of the forest stations had an adequate number of vehicles (lorries, landrovers, tractors). For example, Chogoria Forest Station had no vehicle at all, Castle Station only one tractor, Chehe and Ragati Forest Stations shared a single old landrover, but could not afford petrol. Consequently, the foresters were dependent on public transport to fulfil their duties and on vehicles provided by the local sawmills. Most Forest Stations were not equipped with communication systems, making immediate action against tree-poaching activities impossible. The forest guards, who were paid only a basic salary, had no means of confronting the often heavily armed illegal loggers (6).

There have been no improvements in the situation over the past few years. The pressure on Mount Kenya's forests has increased tremendously and vast areas have been depleted (Fig. 1). However, the overall extension of the Forest Reserve has not changed in any large degree since early this century (7).

The Camphor Forests

Large tracts of the *Ocotea* forests of Mount Kenya were destroyed long before colonial times by Gikuyu, Meru and Embu settlements, and by the Ilpurko and Illaikipiak Maasai (8–11). Only a large bulge west of Chuka on southwest Mount Kenya remained. In colonial times, and even more so since Kenya became independent, these forests underwent enormous change due to intensive selective logging. As a consequence, the vegetation cover has changed to secondary forests, dominated by the Euphorbiaceae *Macaranga kilimandscharica* in large areas of the lower and medium altitudes, and *Neoboutonia macrocalyx* in the higher altitudes. *Ocotea usambarensis*, East African camphor, is logged for its hardwood, which like cedar (*Juniperus procera*), is very resistant to termites. Since camphor is found as individual trees in broadleaved mixed forests, and never forms pure stands, numerous main logging roads and small tracks were established to gain easy access to the trees, and large forest areas were destroyed in the process.

In the Chuka area, the forests have remained relatively untouched because they functioned as a border between rival tribes (11), and due to poor access to the deep valleys. During the 1980s, countless legal and illegal woodcutter camps were established even in these remote forests, irrespective of the presidential ban on cutting indigenous trees in 1986. As the removal of dead wood was still allowed, the usual practice was to fell as many *Ocotea* trees as possible by digging the tree up by the roots, then leaving the tree for some time to collect the "naturally fallen" material later. Especially along the Chogoria route, countless *Ocotea* stems felled in this way, were found slowly decomposing in the forest. As this method destroys all saplings originating from the old roots, no regeneration of camphor is possible.

When *Ocotea* became depleted near the old routes of access (mainly the Chogoria track), these were broadened to allow heavier trucks to go into the forests, destroying all vegetation on the roadsides. The number of smaller tracks also increased significantly. In 1990, old *Ocotea* trees were still frequent, e.g. along Chogoria route, whereas as early as 1992 only a few trees were left on the broadened roadside. Today, camphor trees are felled even on very steep slopes, leading to severe erosion. The commonly practiced method of selective logging in these remote areas was, and still is, highly inefficient, as only a minor portion of the trees cut are utilized. Most of the huge stems are too large to be removed via the access tracks. Therefore, they are cloven with power saws and then removed. The bulk of the unused material is left in the forest.

After 1992, the situation in large parts of the Chuka area deteriorated. As the licence fees for extracting timber are the same as in the 1970s, and the fines for illegal cutting incredibly low the timber trade is yielding very high profits. Tolerated by an overstrained, helpless or corrupt forest administration, large-scale logging was started. Within a few months, several large new access roads were constructed and nearly all camphor trees were removed in some parts of the region, leaving vast stands of totally devastated forest. Whole stems were removed using heavy equipment, which destroyed the remaining vegetation. So far, no legal action has been taken to counteract these activities. Consequently, the remaining more or less undisturbed *Ocotea* forests in Kenya, are highly endangered.

Besides logging, cultivation of marihuana (Banghi) is another very serious threat to the forests on the southeastern slopes of Mount Kenya. From the air, numerous marihuana shambas of all sizes can be seen. These shambas were established deep in the forest by slash and burn activities. Although these activities are illegal, only a few of the people working in the Banghi shambas tried to hide from the circling plane. After drying the harvested material in the forest or sometimes even in the Nyayo Tea Zone, the drug is transported to agents/consumers. Investigations at the Forest Department revealed clearly that the authorities are fully aware of these activities. Members of the KIFCON (Kenya Indigenous Forest Conservation Project) staff were even warned against visiting the area concerned without having an armed escort, as the shambas "are always guarded by heavily armed men". They were also warned against investigating the matter, as "the drug is being transported to the consumer in vehicles of the Kenyan government" (KIFCON). The involvement of even high officials in this Kenyan drug trade and forest destruction is obvious.

The collection of honey does not severely affect the forests, as the forest areas are too wet to lead to fires, which might be kindled by careless honey hunters. Besides which, very little wood is needed for the construction of beehives. From the point of view of the forest administration, the high population density of large herbivores, mainly elephants, was claimed to be responsible for the failure of *Ocotea* regeneration. However, this argument was not corroborated by the data of this study. In ar-



New access road to undisturbed forest areas. Photo: R.W. Bussman.

Marihuana cultivation on southern Mount Kenya after slash-and-burn of the primary forest. Some dead trees are still left standing in the fields. Photo: R.W. Bussmann.



eas with an estimated equally high elephant population (estimated on the basis of dung quantity), regions without heavy logging showed considerable regeneration, whereas in logging areas, especially when the *Ocotea* trees had been felled by uprooting, almost no regeneration was observed. Therefore, it appears that the main reasons for the decreasing regeneration of *Ocotea usambarensis* are overexploitation and unsustainable logging practices.

The Cedar Forests

The submontane and montane cedar forests have been, and still are, heavily affected by human activities. In many areas of the original Cedar forests, plantations of fast growing softwoods (before 1945 mainly *Cupressus lusitanica*, more recently *Pinus radiata* and *P. patula* (4)) have been established, and heavy logging has devastated large parts of the area. Vast forest areas are still used for livestock grazing, which leaves practically no natural vegetation.

In late 1993, after clearfelling, some plantations of *Cupressus lusitanica* and *Pinus radiata* in the Naro Moru and Sirimon regions were opened, by presidential decree, to temporary farm-

ing activities. As a consequence, shambas are now extending almost to the Naro Moru Gate of Mt. Kenya National Park, and the remaining forests are subjected to extreme pressure from squatters felling more trees for timber and firewood, and burning indigenous forest areas to provide arable land (Fig. 6).

On the drier northern slopes of the mountain a so called "forest gap" has been mentioned by several authors (7, 12). Whether this gap is natural or man-made is a matter of debate. According to long-term residents in the area, this gap has been widened by clearing the forest for farming purposes. In the late 1970s, some areas of the Mt. Kenya Forest Reserve were excised by the Kenyan Government for farmland. In many shambas (fields) in this area, forest trees or remnants of trees are still present, indicating the former extension of the forest belt. It is not likely that climatic conditions are responsible for the formation of the gap, since in an even drier area exactly north of the gap on Mt Kenya, there is the Ngare Ndare forest with extensive stands of *Juniperus procera*. Therefore, there is no reason to assume natural causes for the existence of the Mt. Kenya forest gap. Most probably there was a closed forest belt where this gap now exists.

Nevertheless, since the gap was already reported at the turn of the 20th century, it must have been generated a long time ago. Prior to the establishment of the "White Highlands", the area in question was inhabited by the Illaikipiak Maasai (9) who were pastoralists and may have cleared the forest for livestock grazing. Burning shortly before the long rains improves the growth of grass, a method which is still practiced in pastoral communities.

The remaining Cedar forests are heavily affected by legal and illegal logging for timber, especially for fence poles, since the durable, termite-resistant cedarwood is most suitable for this purpose. Even the forests along Sirimon route, which are now part of the Mt. Kenya National Park, have been severely impaired by such activities. After the region became part of the park, illegal cutting was still continued, especially so in the readily accessible parts of the forests.

In wide parts of the *Juniperus* forests, only a few seedlings and young individuals of the dominant tree species were encountered. Therefore, fundamental changes in the composition and physiognomy of these forests must be envisaged for the long term. In contrast to the Camphor forests, the huge game populations, especially buffaloes, contribute much to the destruction of the Cedar forests by feeding on seedlings and young trees. Browsing damage was frequently observed and especially in the dry season, the forest floor is partly bare of vegetation as every herb or bush is eaten and trampled by game. Nevertheless, human activities are mainly responsible for the poor regeneration of the Cedar forest, in particular the too frequent fires caused by arsonists and, occasionally, honey hunters.

The Hagenia Forests

The subalpine Kosso forests are dominated by the large, often horizontally growing *Hagenia abyssinica*. Undisturbed forests of this type show a very open understorey.

Apparently, completely undisturbed *Hagenia* forests were encountered in only few localities on Mt. Kenya. Despite these forests not being impacted by logging activities, severe damage was seen. Most of the forest areas were affected by big game, especially buffaloes, which by frequent trampling on the vegetation create numerous open, grass-covered patches. Debarking of trees by elephants was observed as well as frequent chafing damage on the bark. After the destruction of the natural vegetation, grassy patches are ransacked by armies of mole rats (*Tachoryctes splendens* & *T. rex*), the mounds of which cover thousands of square meters. Forest species cannot regenerate in these areas. However, aggressive moorland species, like *Stoebe kilimand-*



Undisturbed Camphor forests on Eastern Mount Kenya.
Photo: R.W. Bussmann.



Same area after logging started in 1994. Note new access road in the background. Photo: R.W. Bussmann.

scharica, *Euryops brownei* and *Erica arborea* may invade the clearings from the borders, or nitrophilous species like *Rumex ruwenzoriensis* and *Sambucus africana* can establish around places where buffaloes rest. These factors clearly inhibit regeneration of *Hagenia* itself even where, due to previous fires, the Kosso forest could regenerate.

MANAGEMENT

In general, the daily needs of a rapidly growing population have led to destruction of Kenyan forests, and mismanagement and corrupt staff members of the Forest Department and Administration have catalyzed the expansion of destruction unrestrictedly. After many years of fieldwork on Mt. Kenya, it has become obvious that extensive improvements are necessary to protect the remaining forest areas for the sustainable use of further generations. Therefore, the following suggestions for some improvement should urgently be translated into action.

The forest administration must be given a position from which it can truly manage and control sustainable forestry practices. Therefore:

- The salaries of Forest Department staff should be adjusted to levels comparable with those of the Kenya Wildlife Service (KWS).
- Ineffective or corrupt staff members from the departmental and governmental forest administration must be replaced.
- Programs of further education in sustainable forestry and forest ecology should be offered to the staff members of the Forest Department, and be obligatory for higher ranks.

View of Mount Kenya from the tea zone on the southwestern foothills, summit region and alpine zone covered white after heavy snowstorm (14/4, 1993). Photo: R. Bussmann.



- Manpower, transport and communication facilities of Forest Stations and arms of forest guards need to be improved to enable them to fulfil their duties.
- Control gates should be established at all access roads to the forest.

Together with providing the Forest Stations with the necessary equipment, the following steps should be taken to achieve sustainable forest management:

- Complete logging stop for indigenous species, until an inventory of the remaining forests has been completed and a management plan has been agreed on.
- Official marking of all legally extracted timber, so that illegally extracted timber can be confiscated at the gates of the access roads.
- Punishment for illegal activities including transport, processing or sale of illegally logged timber or products out of it with really heavy fines.
- Raising logging license fees, which are still on a level of the 1970s, to a recent level to cut the profits of loggers and to increase the income of the Forest Department.

To allow and support regeneration of the forests, the following measures should be taken:

- Reforestation with indigenous tree species in forest areas affected by logging.
- Obligation for sawmillers to replant indigenous species for every extracted tree.
- Demarcation of "regeneration reserves" where all human activities have to be banned.

Since the main cause of the rapid increase in logging and forest destruction is uncontrolled population growth (from 6 million people in 1963 to more than 26 million in 1993 with more than 50% under 15 yrs of age), steps must be taken to decrease the pressure for firewood, timber and land on Mt. Kenya. Examples of steps include:

- Establishment of tree nurseries and tree planting schemes in the local communities to guarantee their own supply of wood without need to exploit the indigenous forests.
- Eviction of all squatters from the Forest Reserves and ban of any cultivation in the forests.
- A nationwide education program on the importance and ecological role of forests and on forest conservation in schools and communities.

KIFCON (Kenya Indigenous Forest Conservation Project) had begun to deal with several of these issues. Tree inventories in most Forest Reserves have been started, local communities have become involved in conservation of forests; and illegal activities and corruption were revealed. In particular, the latter, and the presidential permission for squatters to move back into the

forests, led to disagreement between KIFCON and some of the officers involved in the Forest Department and administration. For that reason, the work of KIFCON was stopped by the Kenyan Government after an initial period of two years, and most of the data is probably not available to other researchers or to the general public.

At present, the destructional threats to the Kenyan forests are more or less out of control. Large areas of indigenous forest are still cut for timber, burnt for cultivation, and numerous settlements are again to be found in the forest reserve areas; shattering all hopes to preserve the remaining Kenyan forests for further generations. With the current situation, the prognosis of Barnes (2) is likely to come true, if Kenya's policy continues to neglect the ecological consequences of further deforestation, no indigenous forest may remain in 2040.

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13. First submitted 14 February 1995. Accepted for publication 14 June 1995.

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